HITEC UNIVERSITY, TAXILA
DEPARTMENT OF COMPUTER ENGINEERING

IoT Based Smart Robotic Assistance for Alzheimer's Patient

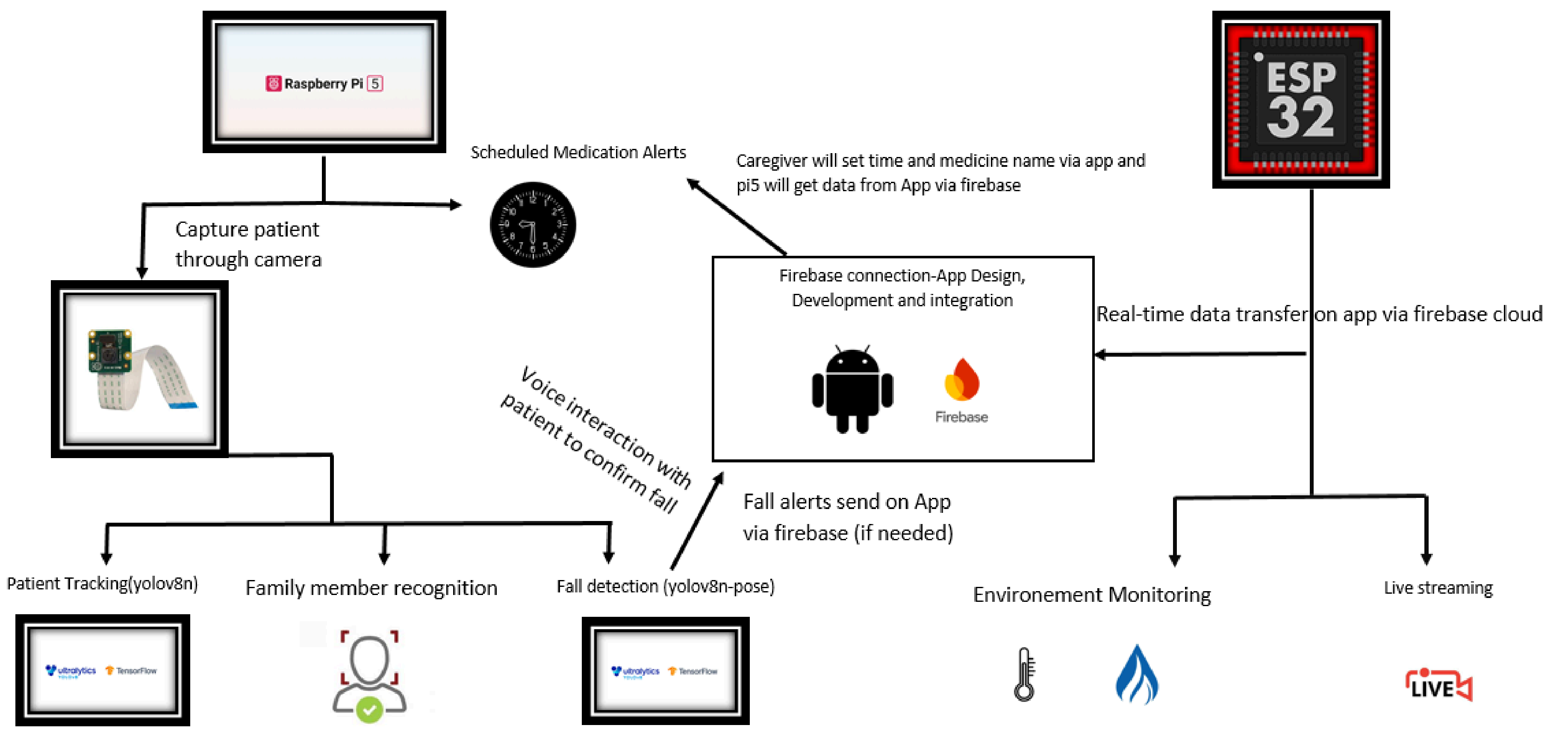
Abstract

The IoT-Based Smart Robotic Assistance System for Alzheimer's Patients is an AI-powered healthcare robot designed to enhance patient safety, monitoring, and caregiver support. The system utilizes computer vision and artificial intelligence to perform real-time fall detection, patient tracking, and family member recognition. It continuously monitors patient activity, detects emergencies, and sends instant alerts to caregivers through Firebase cloud integration and a dedicated mobile application. The solution also incorporates an intelligent medication reminder system that provides timely voice alerts to patients. By combining robotics, AI, IoT, and cloud technologies, the proposed system aims to improve the quality of life, independence, and healthcare management of Alzheimer's patients while reducing caregiver burden.

Methodology

The AI-based elderly care and monitoring system uses a Raspberry Pi 5 as the central processing unit to continuously monitor the elderly person's activities and safety. A Pi Camera captures real-time video, which is processed using YOLOv8 for patient tracking and fall detection. Family member recognition is performed using Haar Cascade for face detection and the LBPH algorithm for face recognition, enabling the system to distinguish authorized family members from strangers.The system also includes voice-based interaction, where speech commands are converted to text using speech recognition and responses are generated through text-to-speech technology. In case of a fall, unknown visitor detection, or emergency situation, the system generates alerts and notifications for caregivers. The complete setup consists of a Raspberry Pi 5, Pi Camera, AI models for tracking and recognition, speech processing modules, and an alert generation mechanism to provide continuous monitoring and assistance for elderly individuals.

Block Diagram


Raspberry Pi 5
Capture patient through camera
Scheduled Medication Alerts
Caregiver will set time and medicine name via app and pi5 will get data from App via firebase
ESP 32
Firebase connection-App Design, Development and integration
Firebase
Real-time data transfer on app via firebase cloud
Voice interaction with patient to confirm fall
Fall alerts send on App via firebase (if needed)
Patient Tracking(yolov8n)
Family member recognition
Fall detection (yolov8n-pose)
Environement Monitoring
Live streaming
ultralytics
TensorFlow
LIVE

Sustainable Development Goal

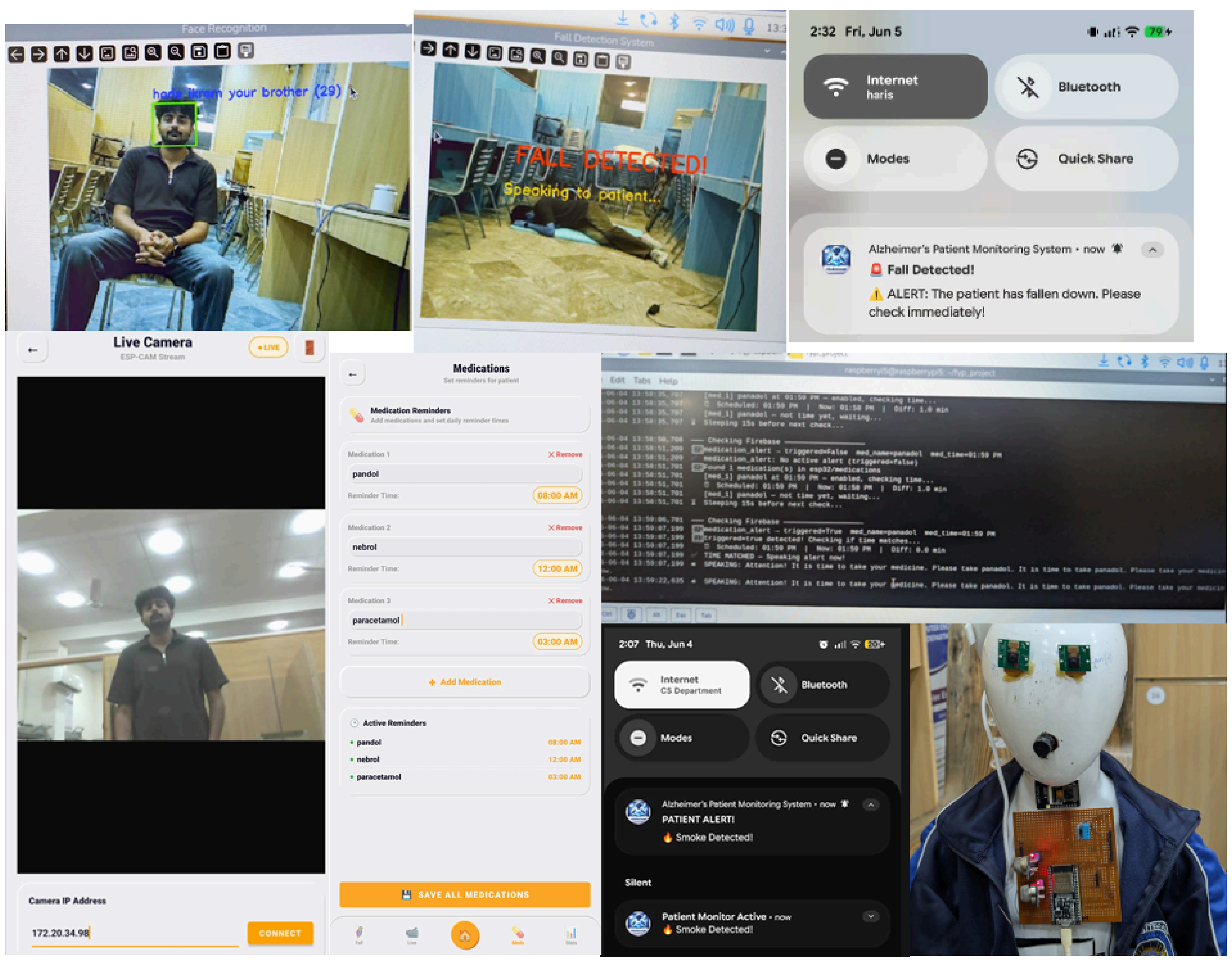
03 GOOD HEALTH AND WELL-BEING

4 QUALITY EDUCATION

09 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Results

- ✓ Successful real-time fall detection using YOLOv8n-based pose estimation with accurate identification of falling posture and sleep posture.
- ✓ Reliable patient tracking achieved through continuous video processing using Raspberry Pi 5 and computer vision techniques.
- ✓ Accurate family member recognition implemented using Haar Cascade for face detection and LBPH algorithm for facial recognition with improved identification performance.
- ✓ Effective medication alert system providing timely reminders and notifications for patients through integrated scheduling and alert mechanisms.
- ✓ Successful environment monitoring with continuous observation of surrounding conditions to enhance safety and detect unusual situations.
- ✓ Functional mobile app integration enabling remote monitoring, alerts, and system control for caregivers in real time.
- ✓ Stable live streaming feature implemented for real-time video transmission from the monitoring system to the user interface with minimal delay.



Alzheimer's Patient Monitoring System • now
🚨 Fall Detected!
⚠️ ALERT: The patient has fallen down. Please check immediately!

Camera IP Address: 172.20.34.98

Medication Reminders
Add medications and set daily reminder times

Medication	Name	Reminder Time
Medication 1	pandol	08:00 AM
Medication 2	nebrol	12:00 AM
Medication 3	paracetamol	03:00 AM

Active Reminders
- pandol 08:00 AM
- nebrol 12:00 AM
- paracetamol 03:00 AM

Alzheimer's Patient Monitoring System • now
PATIENT ALERT!
🔥 Smoke Detected!

Patient Monitor Active • now
🔥 Smoke Detected!

Conclusion

An IoT-Based Smart Robotic Assistance System was developed to enhance the safety, monitoring, and daily support of Alzheimer's patients through AI and smart healthcare technologies. The robot integrates features such as fall detection, patient tracking, family member recognition, medication reminders, and real-time cloud monitoring using Raspberry Pi 5, ESP32, Firebase, and computer vision techniques. The system provides an efficient, intelligent, and cost-effective solution that assists caregivers while promoting patient independence, safety, and improved healthcare management.

Unit Cost:10,54,55

Scan Me

Group Members

Zeenat Fatima	22-ce-002
Muhammad Haris Ikram	22-ce-015
Syed Muhammad Noor uz Zaman	22-ce-026

Supervisor

Dr. Sobia Arhad